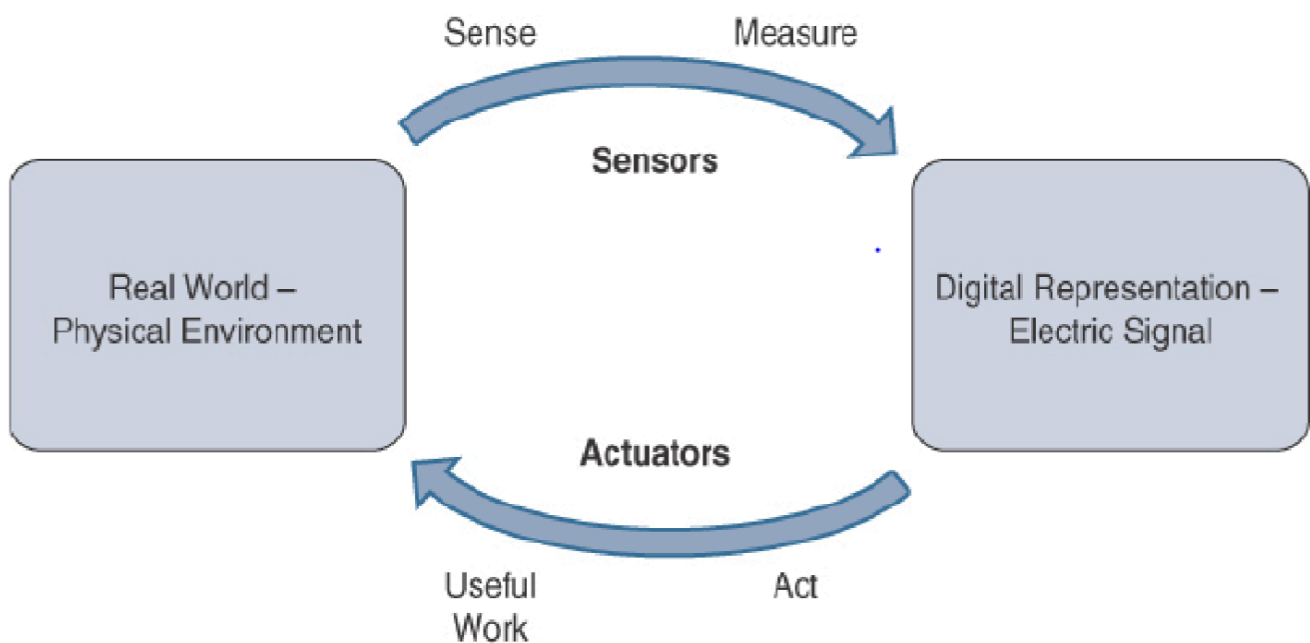


Sensor

1. Active or passive - produce an energy output
 2. Invasive or non-invasive - part of the environment it is measuring
 3. Contact or no-contact - physical contact
 4. Absolute or relative
 5. Area of application
- Sensors convert physical attribute to electrical signal like it sense temperature by measuring heat and converts into electrical signal.
 - Actuator converts electrical signal into physical action by keeping some temperature value as threshold.



Actuators

Classified based on **Type of motion:**

1. Power
 2. Binary or continuous
 3. Area of application
 4. Type of energy
-

Type	Examples
Mechanical actuators	Lever, screw jack, hand crank
Electrical actuators	Thyristor, bipolar transistor, diode
Electromechanical actuators	AC motor, DC motor, step motor
Electromagnetic actuators	Electromagnet, linear solenoid
Hydraulic and pneumatic actuators	Hydraulic cylinder, pneumatic cylinder, piston, pressure control valves, air motors
Smart material actuators (includes thermal and magnetic actuators)	Shape memory alloy (SMA), ion exchange fluid, magnetostrictive material, bimetallic strip, piezoelectric bimorph
Micro- and nanoactuators	Electrostatic motor, microvalve, comb drive

MEMS-Micro-Electro-Mechanical system

MEMS are micro machines can integrate and combine electric and mechanical elements such as machines. It is a technology used to do small integrated devices or systems that is a combination of mechanical and electrical components

Automobiles were among the first to commercially introduce MEMS into the mass market, with airbag accelerometers.

Smart objects

A smart object is an object that enhances the interaction with not only people but also with other smart objects.

Characteristics:

1. Processing unit
2. Sensors and actuators
3. Communication device
4. Power source

Trends:

1. Size is decreasing
2. Power consumption is decreasing
3. Processing power is increasing

M2M (Machine-to-Machine)

Communication or exchange of data between two or more machines without human interfacing or interaction.

The M2M uses a device such as sensor, RFID, meter, etc. to capture an 'events' like temperature, inventory level, etc., which is relayed through a network i.e., wireless, wired or hybrid to an application (software program), that translates the captured event into meaningful information.

An M2M area network comprises of machines (or M2M nodes) which have embedded hardware modules for sensing, actuation, and communication.

Various communication protocols can be used for M2M local area networks such as ZigBee, Bluetooth, ModBus, M-Bus, Wireless M-Bus, Power Line Communication (PLC), 6LoWPAN, IEEE 802.15.4, etc

The communication network provides connectivity to remote M2M area networks. The communication network can use either wired or wireless networks (IP- based). While the M2M area networks use either proprietary or non-IP based communication protocols, the communication network uses IP-based networks.

M2M System Architecture (Components):

1. M2M area networks
2. Communication networks
3. Application domains
4. M2M gateways

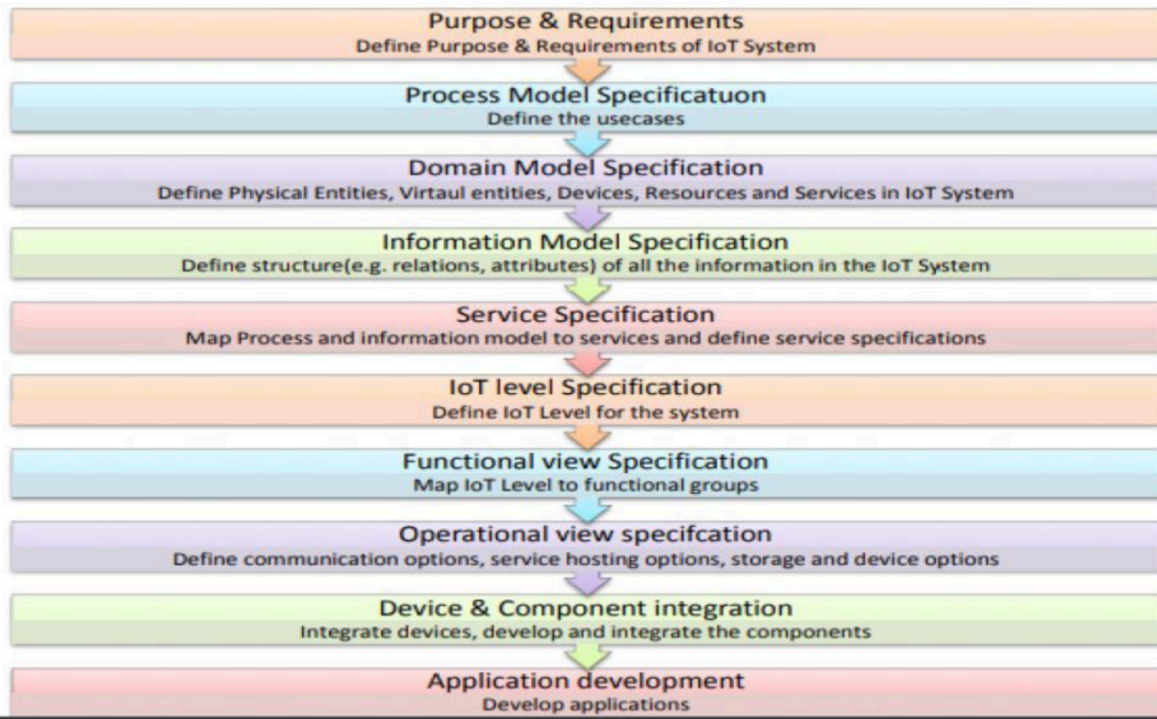
Feature	M2M (Machine-to-Machine)	IoT (Internet of Things)
Communication Protocols	Uses proprietary or non-IP based protocols within its network.	Uses IP-based protocols to communicate over the internet.
Connected Devices	Typically involves homogeneous types of machines within a specific network.	Involves a wide variety of "Things"—physical objects with unique identifiers that can sense and communicate.
Primary Emphasis	More emphasis on the hardware with embedded modules.	More emphasis on the software for data analysis and application services.

Feature	M2M (Machine-to-Machine)	IoT (Internet of Things)
Data Collection	Data is collected in point-to-point solutions, often stored in on-premises infrastructure.	Data is collected and aggregated in the cloud (public, private, or hybrid).
Applications	Data is typically used by on-premises applications like service management and diagnosis tools.	Data is accessed by cloud-based applications for analytics, remote management, and enterprise use.

Difference between IoT and M2M	
IoT	M2M
Digital connectivity among various devices to communicate	Devices connected to the formwork cycle using various machines and devices
Random work, work instructions are given among devices	The action triggered events among devices
Interchange of data is huge as it involves devices, machines, people, things, etc	Cloud computing helps in interacting/data exchange
Sensor integrated devices to enable IoT connectivity	Wired, wireless, cellular, etc
Two-way communication offers an option to all the devices	Mostly one way, based on triggered actions
Solution managing all the connections offer unlimited integration	Requires particular communication rules, resulting in minimal integration
Need internet for most of the cases	No need to rely on the internet

IoT System Design Methodology

Steps involved in IoT System Design Methodology



Advantages of Using Design methodology

- Reducing the design, testing and maintenance time
- Provide better interoperability
- Reduce the complexity

Step 1: Purpose & Requirements Specification

The first step in IoT system design methodology is to define the purpose and requirements of the system. In this step, the system purpose, behavior and requirements (such as data collection requirements, data analysis requirements, system management requirements, data privacy and security requirements, user interface requirements, ...) are captured

Step 2: Process Specification

The second step in the IoT design methodology is to define the process specification. In this step, the use cases of the IoT system are formally described based on and derived from the purpose and requirement specifications.

Step 3 : Domain Model Specification

The domain model describes the main concepts, entities and objects in the domain of IoT system to be designed. Domain model defines the attributes of the objects and relationships between objects. Domain model provides an abstract representation of the concepts, objects and entities in the IoT domain, independent of any specific technology or platform. With the domain model, the IoT system designers can get an understanding of the IoT domain for which the system is to be designed.

Step 4 : Information Model Specification

Information Model defines the structure of all the information in the IoT system, for example, attributes of Virtual Entities, relations, etc. Information model does not describe the specifics of how the information is represented or stored. To define the information model, we first list the Virtual Entities defined in the Domain Model. Information model adds more details to the Virtual Entities by defining their attributes and relations.

Step 5 : Service Specification

Service specifications define the services in the IoT system, service types, service inputs/output, service endpoints, service schedules, service preconditions and service effects.

Step 6: IoT Level Specification

Step 7: Functional View Specification

The Functional View (FV) defines the functions of the IoT systems grouped into various Functional Groups (FGs). Each Functional Group either provides functionalities for interacting with instances of concepts defined in the Domain Model or provides information related to these concepts.

Step 8: Operational View Specification

Various options pertaining to the IoT system deployment and operation are defined, such as, service hosting options, storage options, device options, application hosting options, etc

Step 9: Device & Component Integration

The integration of the devices and components

Step 10: Application Development

The final step in the IoT design methodology is to develop the IoT application.

Challenges in Wireless Medium Communication:

1. Scalability
2. Heterogeneity
3. Reliability and Availability
4. QoS
5. Energy Efficiency - the number of bits that can be sent over a unit of power consumption which is usually quantified by bits per Joule

Protocol	Optimized for Extended Battery Life	Nominal Range Limit	Typical Data Rate	Spectrum
 Bluetooth®	✓	Personal (<10m)	2Mbps	ISM 2.4GHz unlicensed
 NFC	✓	Contact (<4cm)	100kbps	ISM 13.56MHz unlicensed
 WiFi™	✗	Local (<100m)	>100Mbps	ISM 2.4GHz/5GHz unlicensed
 LoRaWAN™	✓	Metro (>10km)	<50kbps	ISM 900MHz unlicensed
NB-IoT	✓	Metro (>10km)	200kbps	Licensed cellular
2G 3G	✗	Metro (>30km)	<2Mbps	Licensed cellular
 4G LTE	✗	Metro (>30km)	>100Mbps	Licensed cellular

Bluetooth:

Benefits:

1. Low Power Consumption
2. Easily upgradable
3. Low Cost of equipment

Issues:

1. Low bandwidth
2. Short Range
3. Security Risk as it leaves data vulnerable to interception

Wi-Fi:

Benefits:

1. Low cost
2. Commonly available
3. Simplicity – No need for extra hardware

Issues:

1. Wi-Fi is not designed to create mesh networks
2. High Power Usage
3. Limited number of devices can be connected

ZigBee:

Benifits:

1. It is less complex than Bluetooth
2. Zigbee has a mesh network topology with low cost, multi hope data transmission
and is power effective
3. Long battery life
4. Supports many nodes

Issues:

1. Short range
2. Low complexity, and low data speed
3. High maintenance cost
4. Replacement with Zigbee compliant appliances can be costly
5. Zigbee is not secure like WiFi based secured system

ZigBee slides:

Zigbee is high-level communication protocols used to create personal area networks with small, low-power digital radios. igbee is a low-power, low data rate, and close proximity (i.e., personal area) wireless ad hoc network.

it is designed to meet low cost battery operated applications. Zigbee standard is specially build for control and sensor network

ZigBee protocol:

It is built on top of IEEE 802.15.4 physical layer

Low data rates of 250 kbps

Distance around 20 to 50 mts

Operates in mesh topoplogy

Using wireless technology reduces costs and eliminates the hassle of running wires.

Zigbee working

1. coordinator - acts as a bridge and root for the entire network
2. router - intermediary devices allowing the data to pass to and from via them to other devices
3. end device - smart thermostat, television, door, CCTV cameras